Online Metastability Detection for Intraday Markets

1 Project Overview

This project studies whether short-horizon market dynamics can be described by a changing local potential landscape, and whether features derived from that landscape improve forecasting.

The pipeline estimates a nonparametric local drift and diffusion from rolling windows of high-frequency price data, reconstructs an implied potential $U(x)$, labels each window as `single_well`, `double_well`, or `unclear`, and then tests whether those regime features improve short-horizon prediction tasks.

The current public version uses synthetic high-frequency data to validate the estimation and modeling pipeline. This choice was practical: the most relevant real high-frequency datasets either require paid access, additional API setup, or more data-engineering infrastructure for reliable ingestion than was available for this initial prototype. The goal of this stage was to validate the end-to-end research workflow first, then move to real market data in a later version. The purpose of this stage is to confirm that the workflow behaves sensibly, produces interpretable structural outputs, and supports downstream modeling before applying it to real market data.

The current notebook run should therefore be interpreted as a **research prototype and pipeline validation**, not as a final trading result.

2 Research Question

Instead of asking only whether lagged returns or liquidity predict the next move, this project asks:

> Can we estimate the local dynamical structure of the market itself, and use that structure as a predictive feature?

For each rolling window, the code:

1. computes log-price $x_t = \log P_t$,
2. estimates conditional drift and diffusion,

3. reconstructs the implied potential

$$U(x) = -\int \mu(x)\, dx,$$

4. extracts structural features such as number of wells, barrier height, well distance, well asymmetry, drift magnitude, and diffusion level,
5. and uses those features in forecasting models alongside standard baseline features.

The interpretation is straightforward:

- a `single_well` potential suggests one dominant local equilibrium,
- a `double_well` potential suggests two competing local states,
- an `unclear` potential suggests weak or unstable structure.

3 Data

The current public version uses synthetic high-frequency data to validate the estimation and modeling pipeline.

This setup is useful because it allows the project to test whether:

- rolling drift and diffusion estimation works,
- the implied potential can be reconstructed cleanly,
- regime labels are stable and interpretable,
- and the downstream machine-learning pipeline runs end to end.

At this stage, the synthetic run is serving as a **research sandbox**, not as evidence of real-market alpha or live trading performance.

The intended next step is to apply the same framework to real high-frequency datasets such as Uniswap v3 pool data, centralized limit-order-book data, or other trade-level intraday datasets.

4 Prediction Tasks

The current notebook evaluates two tasks:

4.1 Future realized volatility

Predict realized volatility over the next fixed horizon.

4.2 Continuation vs reversal

Predict whether short-horizon direction continues or reverses relative to recent momentum.

These are intentionally simple first targets. They allow us to test whether potential-based regime features add anything beyond lagged returns, liquidity, and flow variables.

5 Current Run Summary

5.1 Regime distribution

The estimated regime counts are:

Regime	Count
`single_well`	580
`unclear`	102
`double_well`	7

This means the estimated dynamics are overwhelmingly classified as `single_well`, with occasional unclear windows and very few `double_well` detections.

5.2 Model comparison

Task	Model	RMSE	R^2	Accuracy	AUC
Future vol regression	baseline	0.001747	-0.579066	–	–
Future vol regression	baseline + potential	0.001751	-0.584716	–	–
Continuation classification	baseline	–	–	0.565217	0.603415
Continuation classification	baseline + potential	–	–	0.579710	0.599610

6 Interpretation of Results

6.1 The structural estimation pipeline works

The project successfully completes the full workflow:

- rolling drift and diffusion estimation,
- implied potential reconstruction,
- regime classification,
- structural feature extraction,
- and downstream prediction.

That is important because the first challenge in a project like this is getting the structural estimation pipeline to produce plausible and stable outputs.

6.2 The current run is dominated by single-well structure

Most windows are classified as `single_well`. The potential snapshots are smooth bowl-shaped curves, which suggests the estimator is mostly identifying one dominant local equilibrium rather than a strong two-state structure.

This is a meaningful result as it implies that under the current settings, metastability is not the dominant regime.

6.3 Double-well detection is too rare to drive the results

Only 7 windows are labeled `double_well`. That is too sparse to support a strong claim that metastable dynamics are a major predictive driver in this run.

This is likely one reason the potential-based feature set does not materially improve the volatility model.

6.4 The volatility model is not yet strong

Both regression models have negative R^2, which means they do not outperform a simple mean-style benchmark in a meaningful way on this run. The predicted volatility series is also smoother than the realized series and misses several spikes.

So while the structural estimation is functioning, the forecasting layer is not yet producing a strong volatility result.

6.5 Potential features add only weak incremental value

The full model does not improve volatility prediction: RMSE is marginally worse than the baseline.

For continuation classification, the full model slightly improves accuracy from 0.565 to 0.580, but AUC remains essentially unchanged and slightly lower. That suggests the additional features may affect the thresholded classification decision a little, but they do not create a clearly better ranking signal.

6.6 Liquidity dominates the regression model

The feature importance chart shows that `past_liquidity_mean` dominates the regression model by a wide margin. Structural features such as `diffusion_std`, `drift_abs_mean`, `current_drift`, and `current_diffusion` do appear, but their contribution is much smaller.

This suggests that the current setup is still behaving primarily like a liquidity-state model with structural overlays, rather than a regime-driven predictor.

7 Figures

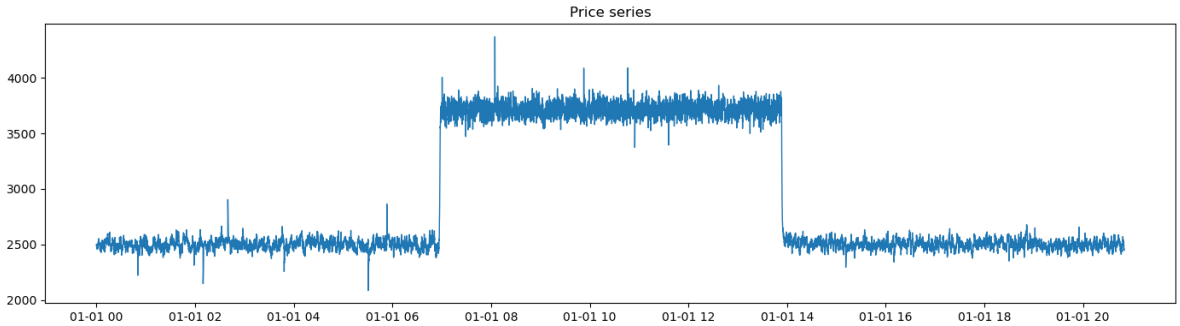

Figure 1: Price series from the synthetic-data validation run. The chart shows broad level shifts that provide a useful testbed for regime estimation.

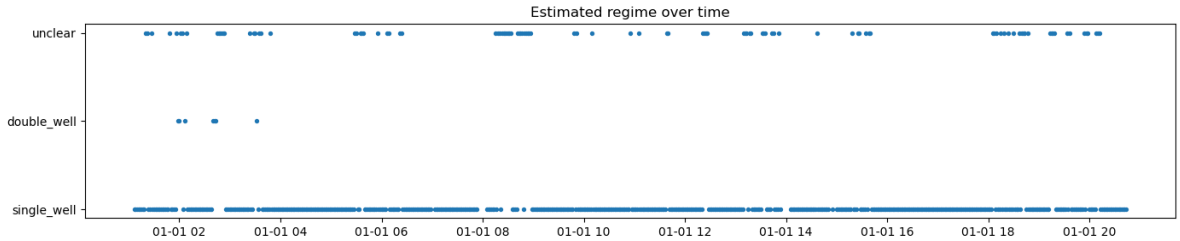

Figure 2: Estimated regime over time for the synthetic-data validation run. The timeline is dominated by `single_well` classifications, with occasional `unclear` windows and only a few `double_well` detections.

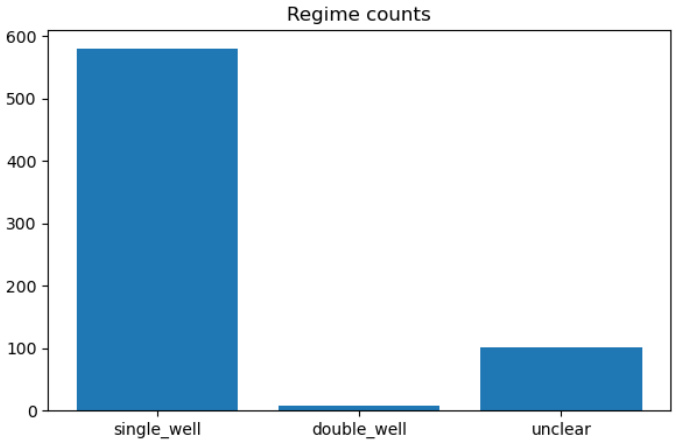

Figure 3: Regime counts from the synthetic-data run. The strong imbalance toward `single_well` is one of the central findings of the current prototype.

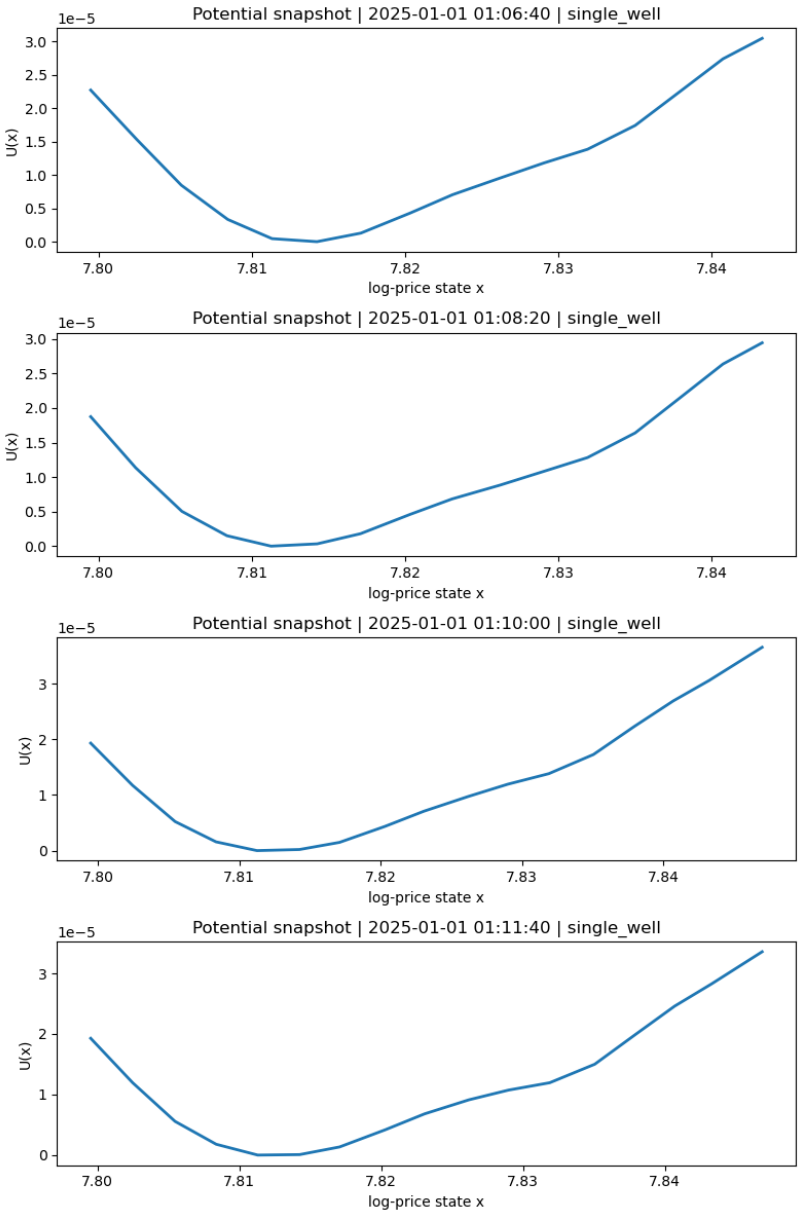

Figure 4: Example potential snapshots from the synthetic-data validation run. These are visually clean and mostly convex, supporting the interpretation that the current run is dominated by single-well structure.

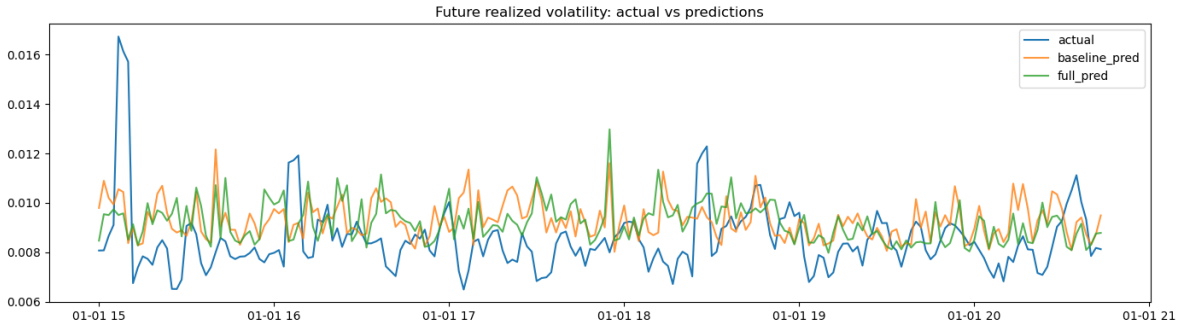

Figure 5: Future realized volatility: actual values versus model predictions in the synthetic-data run. Both models track the average level but miss several realized spikes, and the full model looks very similar to the baseline.

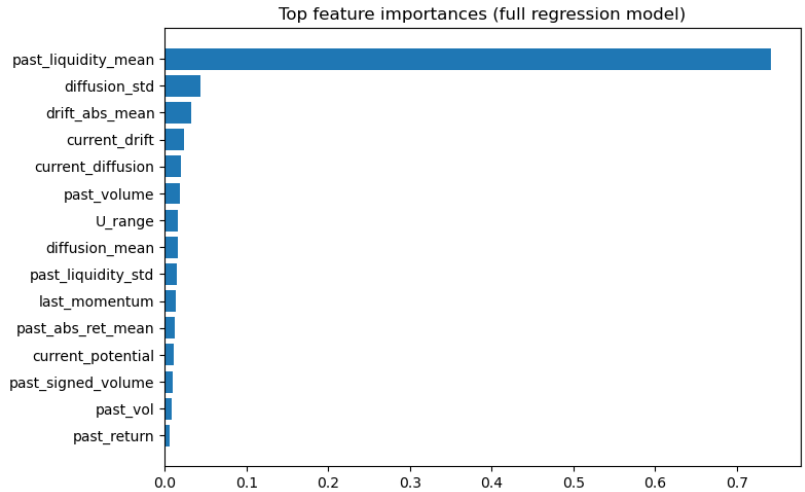

Figure 6: Top feature importances from the full regression model in the synthetic-data run. The dominant feature is `past_liquidity_mean`, while structural features contribute only secondarily.

8 What the Current Run Means

This is a **good prototype** but not yet a strong final result.

The current run supports the claim that:

> the pipeline can estimate and use potential-based regime features in a stable and interpretable way.

It does **not yet** support the stronger claim that:

> metastability-based features clearly improve short-horizon forecasting.

Therefore, the project has moved beyond the stage of being only an interesting idea, but it still needs stronger empirical lift before it becomes a standout quantitative research result.

9 Main Limitations

The current version has several clear limitations:

- **Synthetic-data evaluation only:** the current public results are based on synthetic data used for pipeline validation, not real market data.
- **Regime imbalance:** double-well states are too rare.
- **Weak predictive uplift:** the potential features do not materially improve the volatility model.
- **Simple model class:** random forests are convenient for prototyping but are not necessarily ideal for structured time-series prediction.
- **Basic validation design:** the current split is a simple time split rather than a more careful walk-forward framework.
- **State representation is still limited:** the strongest signal appears to come from liquidity, not metastability.

10 Recommended Next Steps

The clearest next improvements are:

10.1 Improve the state representation

Add richer inputs such as order-flow imbalance, local liquidity shape, jump indicators, regime duration, and transition features.

10.2 Make regime detection less brittle

Experiment with different window sizes, bin counts, smoothing choices, and double-well thresholds.

10.3 Strengthen evaluation

Use walk-forward validation, stronger baselines, horizon sensitivity checks, and calibration analysis.

10.4 Move to real data

Run the same framework on Uniswap v3 data, centralized order-book data, or other high-frequency trade-level datasets.

10.5 Use a more economic target

A better next target would be adverse selection, passive-fill toxicity, reversal probability after a fill, or liquidity deterioration.

11 Conclusion

The most important outcome of the current run is not raw predictive performance. It is that the project now has a functioning end-to-end research pipeline:

- rolling structural estimation,
- implied potential reconstruction,
- regime classification,
- feature extraction,
- downstream prediction,
- and result visualization.

The current evidence suggests that the structural features are plausible and partially informative, but not yet strong enough to carry the forecasting task on their own.

Simply put:

> This project has reached the working research prototype stage, but it still needs real-data validation, stronger regime detection, and stronger empirical lift to become a standout quant result.